

# Adversarial Networks for Data Generation and Domain Adaptation

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A gentle introduction

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CALGARY

# Outline

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- Learning Goals
- Introduction
- Generative Adversarial Networks (GANs)
- Domain Shift and Domain Adaptation
- Domain Adversarial Neural Networks
- Summary

# Learning Goals

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- Introduce Generative Adversarial Networks
  - What they do
  - How they work
- Introduce domain shift and domain adaptation concepts
  - Introduce domain adversarial models for domain adaptation

# Generative Adversarial Networks (GANs)

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# Introduction

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- GANs are unsupervised deep learning methods
- GANs are considered one of the greatest deep learning breakthroughs in recent years
- There are many types of GANs
  - Wasserstein GAN
  - Cycle-GAN
- They all operate under the same principle of having modules with adversarial (*i.e.*, competing objectives)

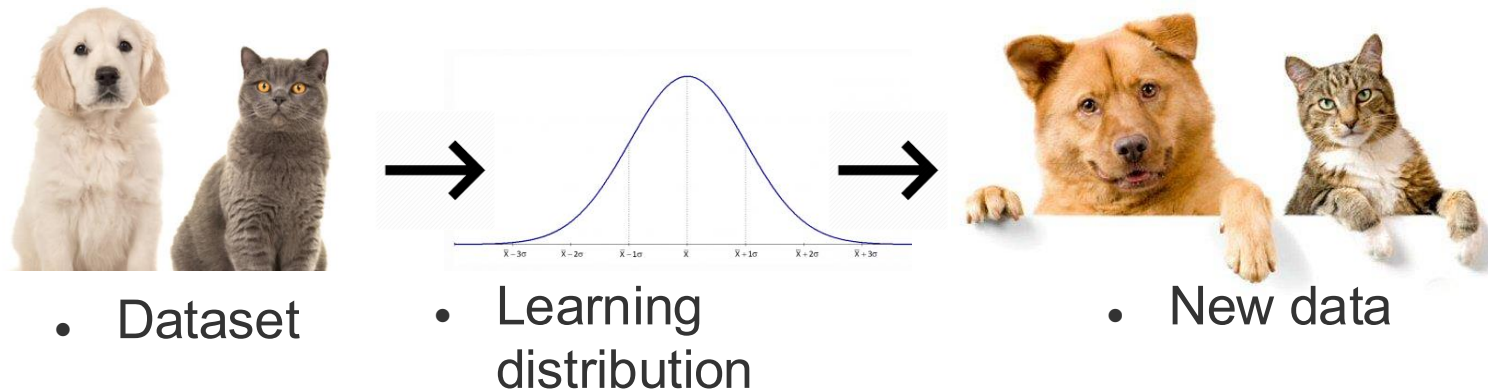
# What are GANs?

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- GANs are generative models. What is a generative model?
- We have two different models in machine learning:
  - 1) Discriminative models

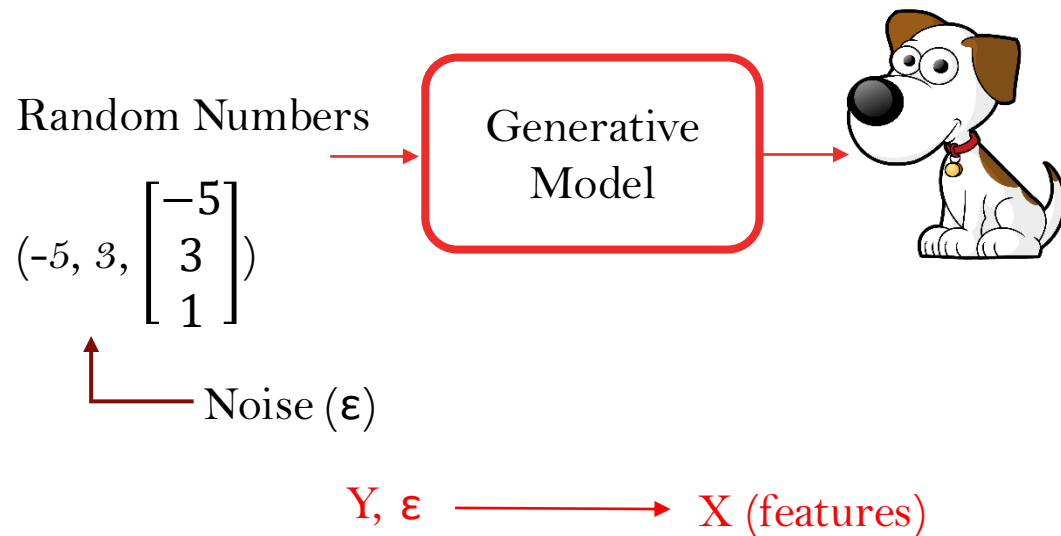


- 2) Generative models



# Machine Learning Models

- Generative Models
  - Generate realistic representation for each class.

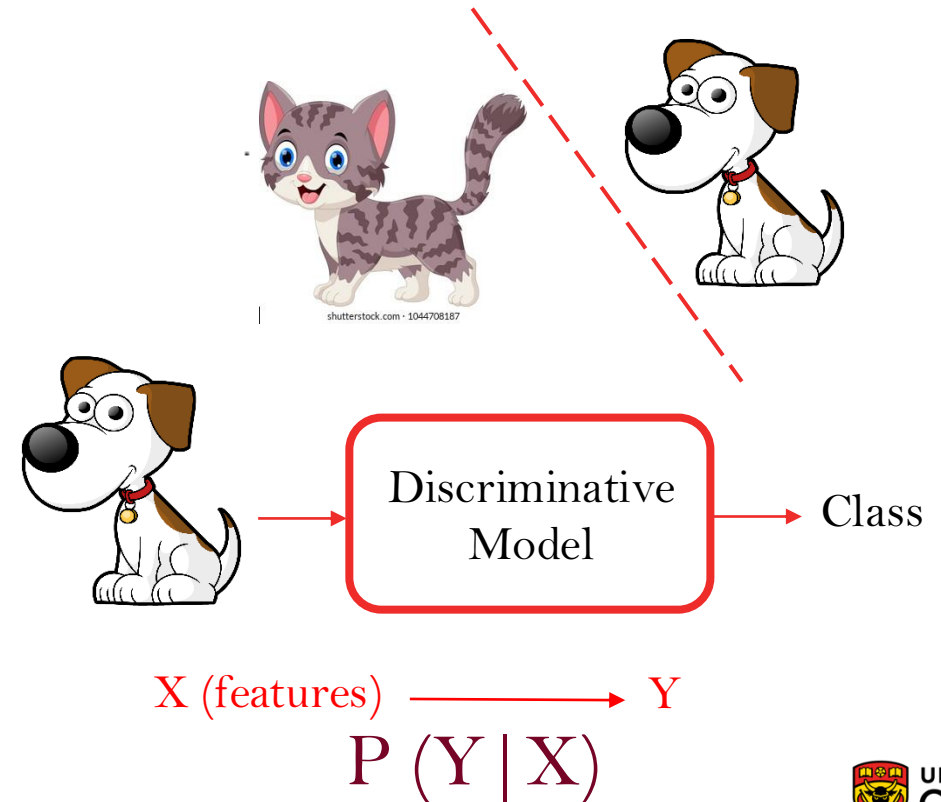


$$P(X | Y)$$

or

$$P(X)$$

- Discriminative Models
  - Used for classification problem



# Generative Adversarial Networks

To produce Realistic Presentation of different classes

To distinguish real images from fake ones (produced by generator)

$$\begin{bmatrix} -5 \\ 3 \\ 1 \end{bmatrix}$$

Generative Model



Discriminative Model

Fake/ Real

There is a competition here !

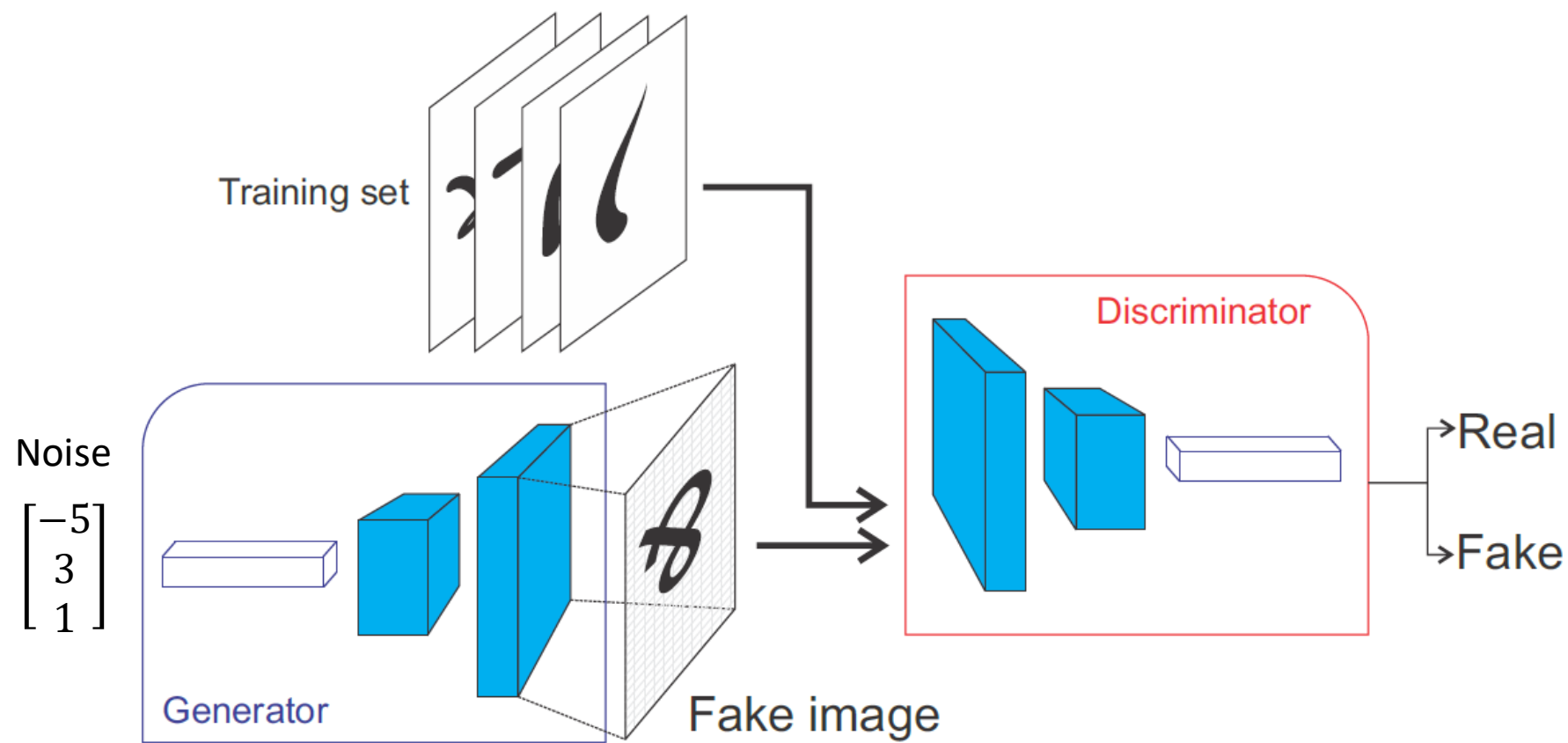
Generator tries to make fakes that look real  
and fool the discriminator

Discriminator learns how to distinguish  
reals from fakes

**GANs are generative  
models where the data  
distribution is learned  
implicitly!**

# GAN

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# GANs Problems

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- **Non-convergence:** the model parameters oscillate and the model does not converge
- **Mode collapse:** the generator collapses and produces a limited number of different samples
- **Diminished gradient:** the discriminator is too good that the generator gradient vanishes and learns nothing,
- **Highly sensitive** to the hyperparameter selections.

# Summary

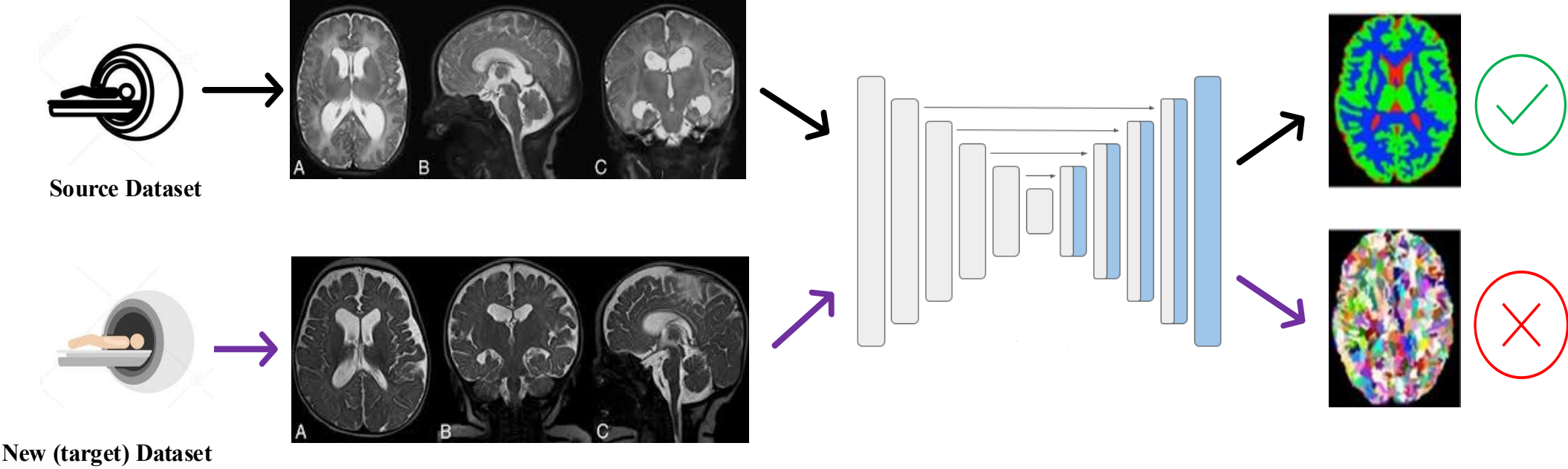
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- GANs are unsupervised techniques
- They can be used to generate synthetic data that can potentially be used to train other deep learning models
- There are different GAN types, but they are all based on the principle of having two competing objectives
- GANs often face instabilities during training

# Domain Shift and Domain Adaptation

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# Motivation

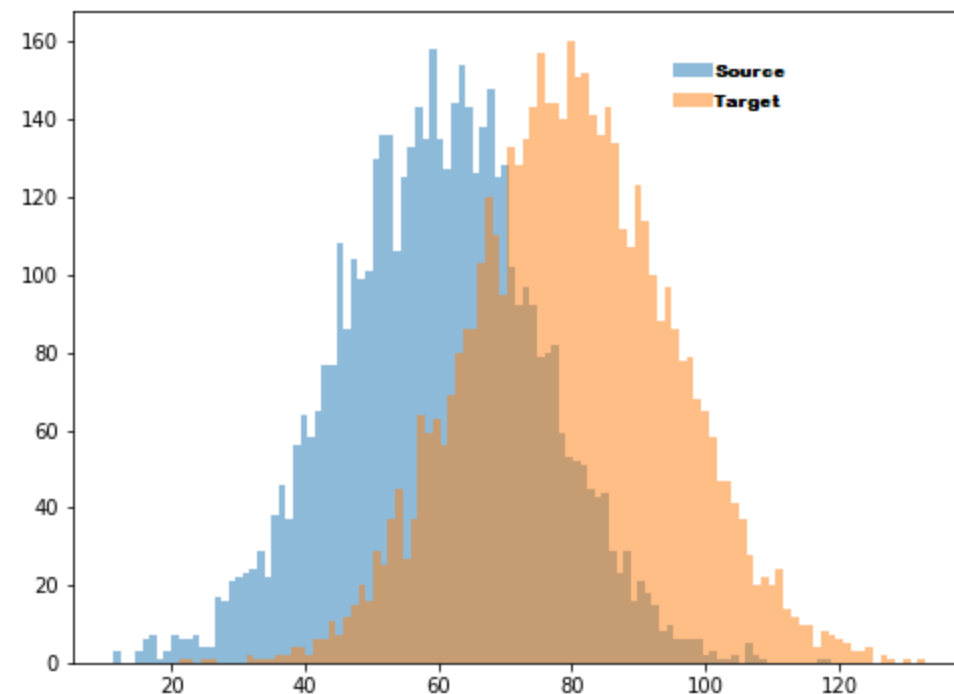


# Domain Shift

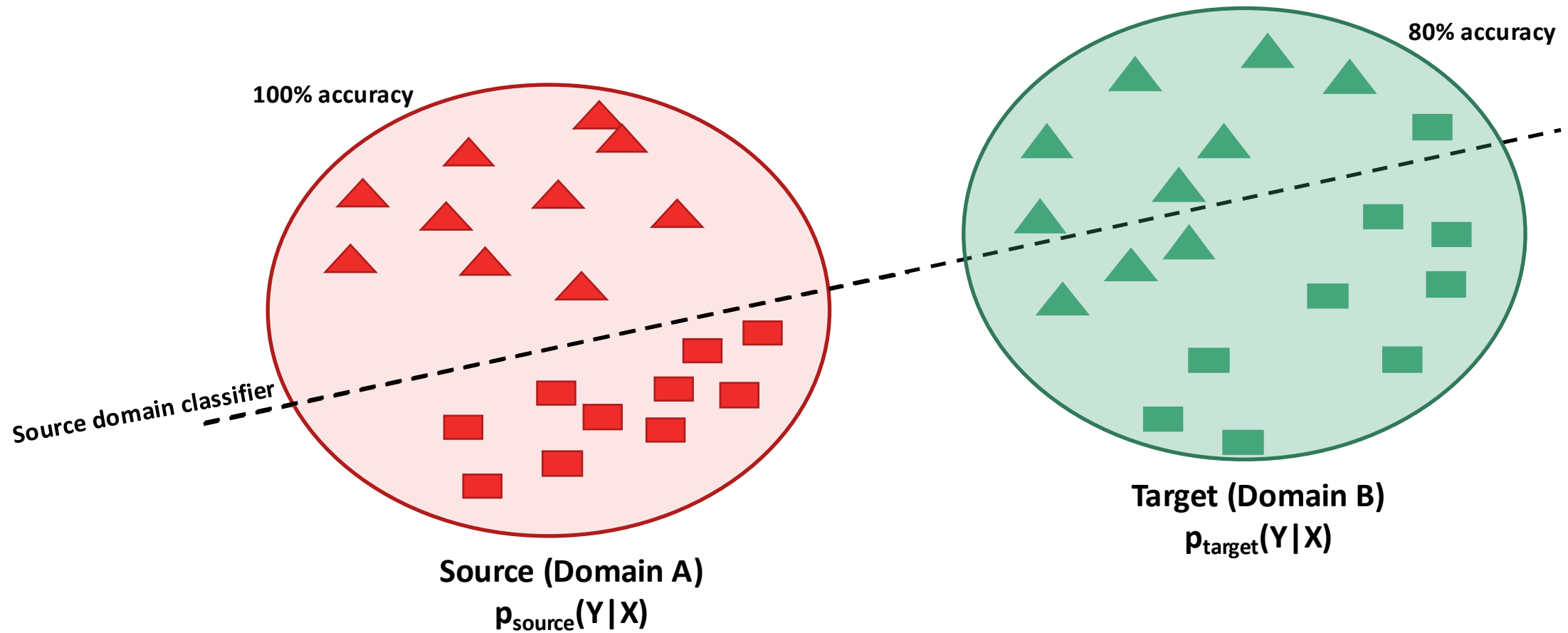
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- **Domain shift:** refers to the change of data distribution between one dataset (source/reference domain) and another dataset (target domain).

$$p_{\text{source}}(Y|X) \neq p_{\text{target}}(Y|X)$$



# Domain Shift Problem



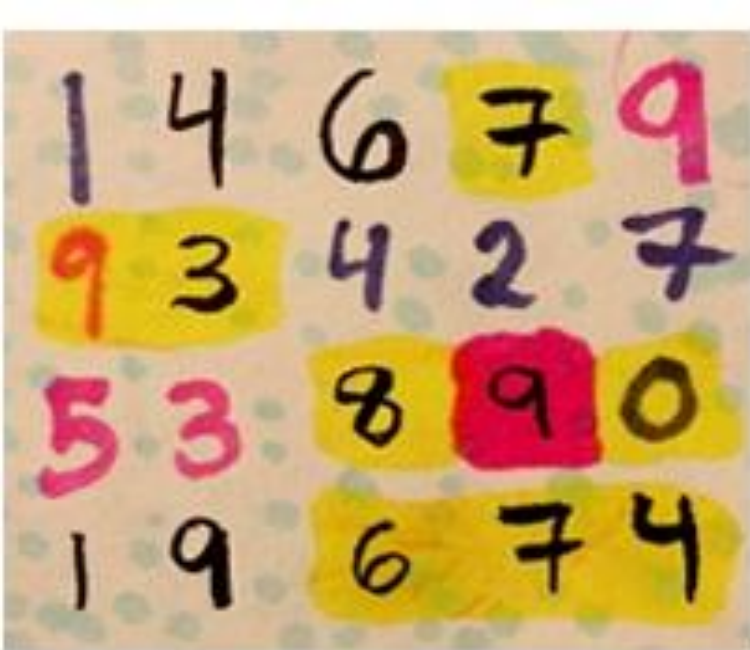
# Different Types of Images

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Domain A

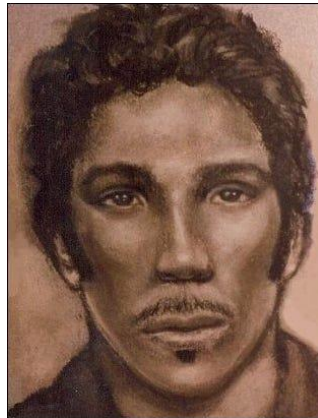


Domain B

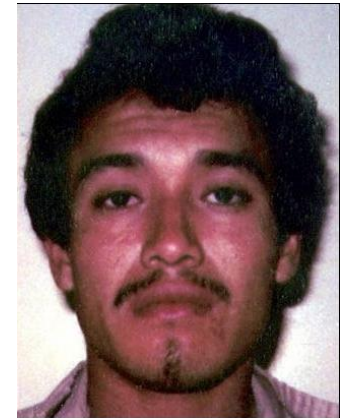


# Different Types of Images: Sketches and Photos

**Domain A: Sketches**

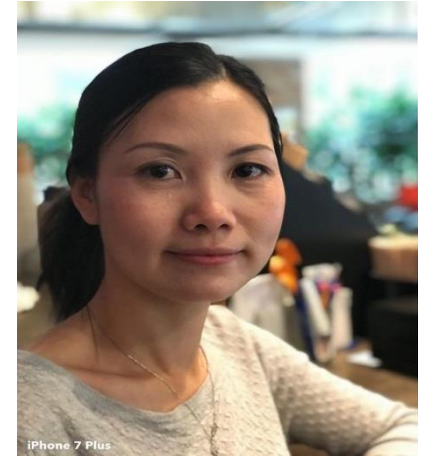


**Domain B: Photos**



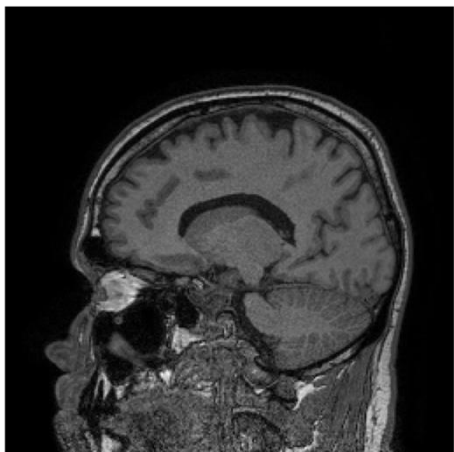
# Technology Differences and Evolution

Camera comparison images: Low light/night mode

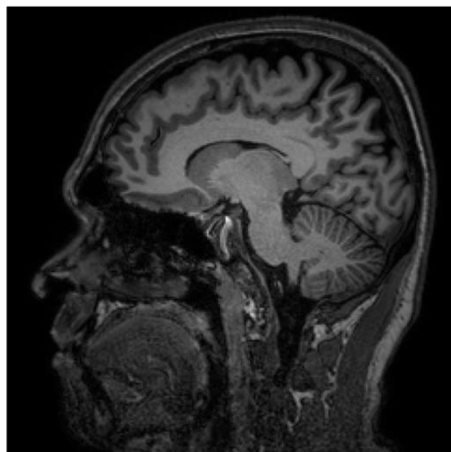


# Hardware and Software Differences

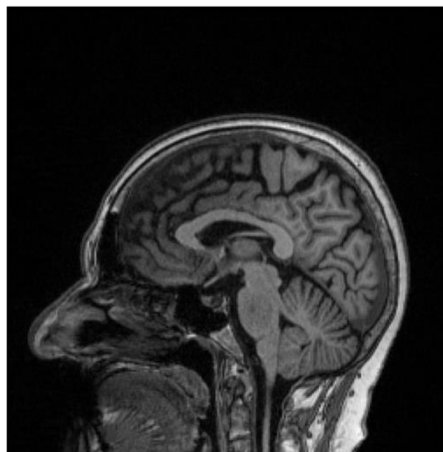
philips\_15



philips\_3



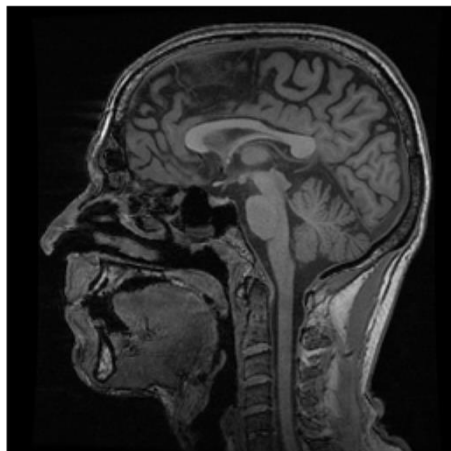
siemens\_15



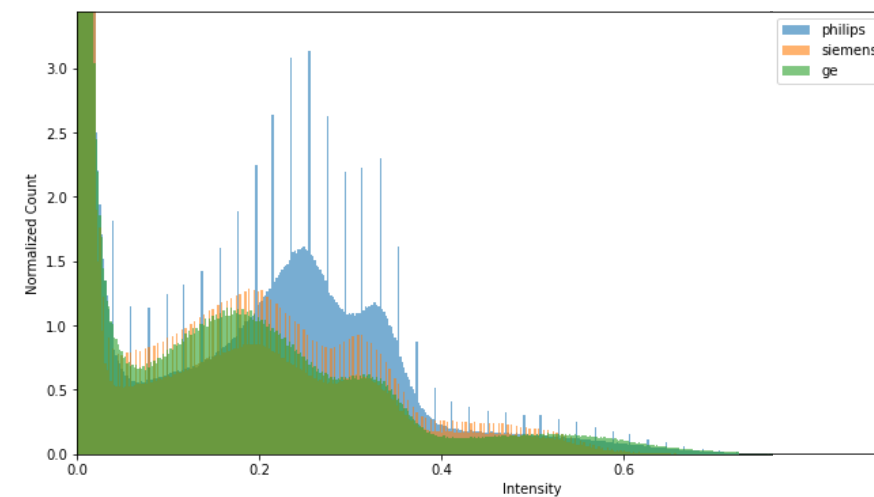
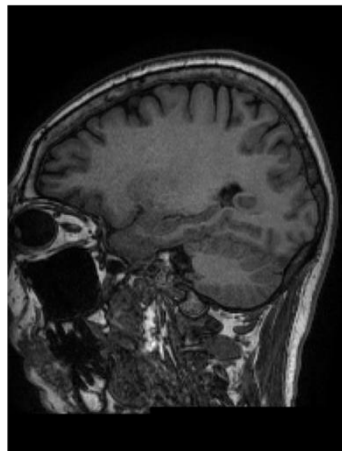
siemens\_3



ge\_3



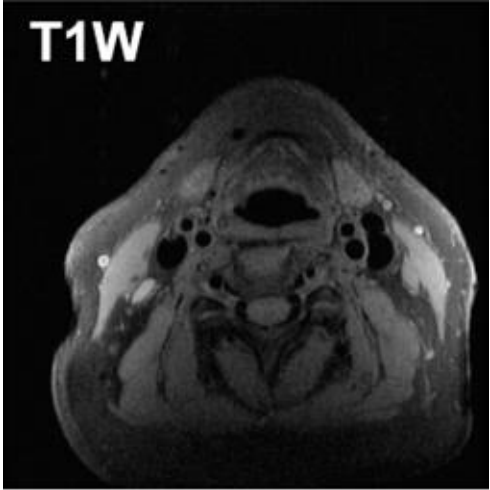
ge\_15



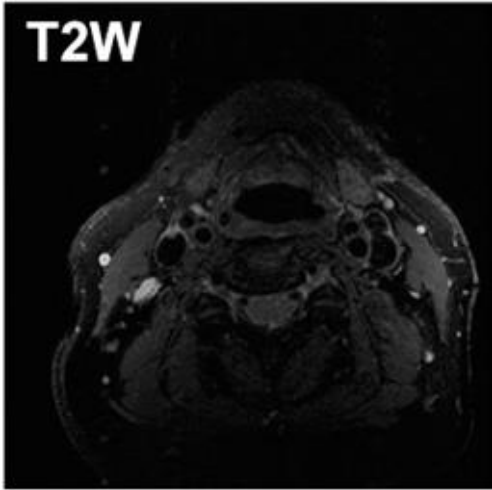
# Hardware and Software Differences

AIM-HIGH Study

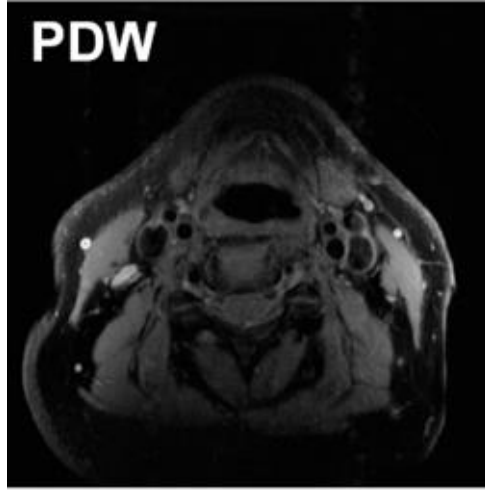
T1W



T2W



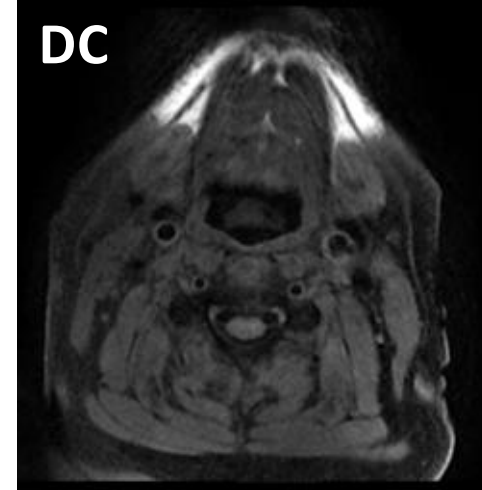
PDW



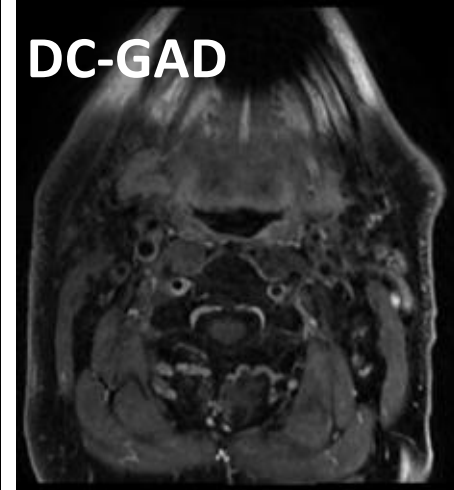
- The carotid arteries were manually annotated at the time of the study

CARDIS Study

DC



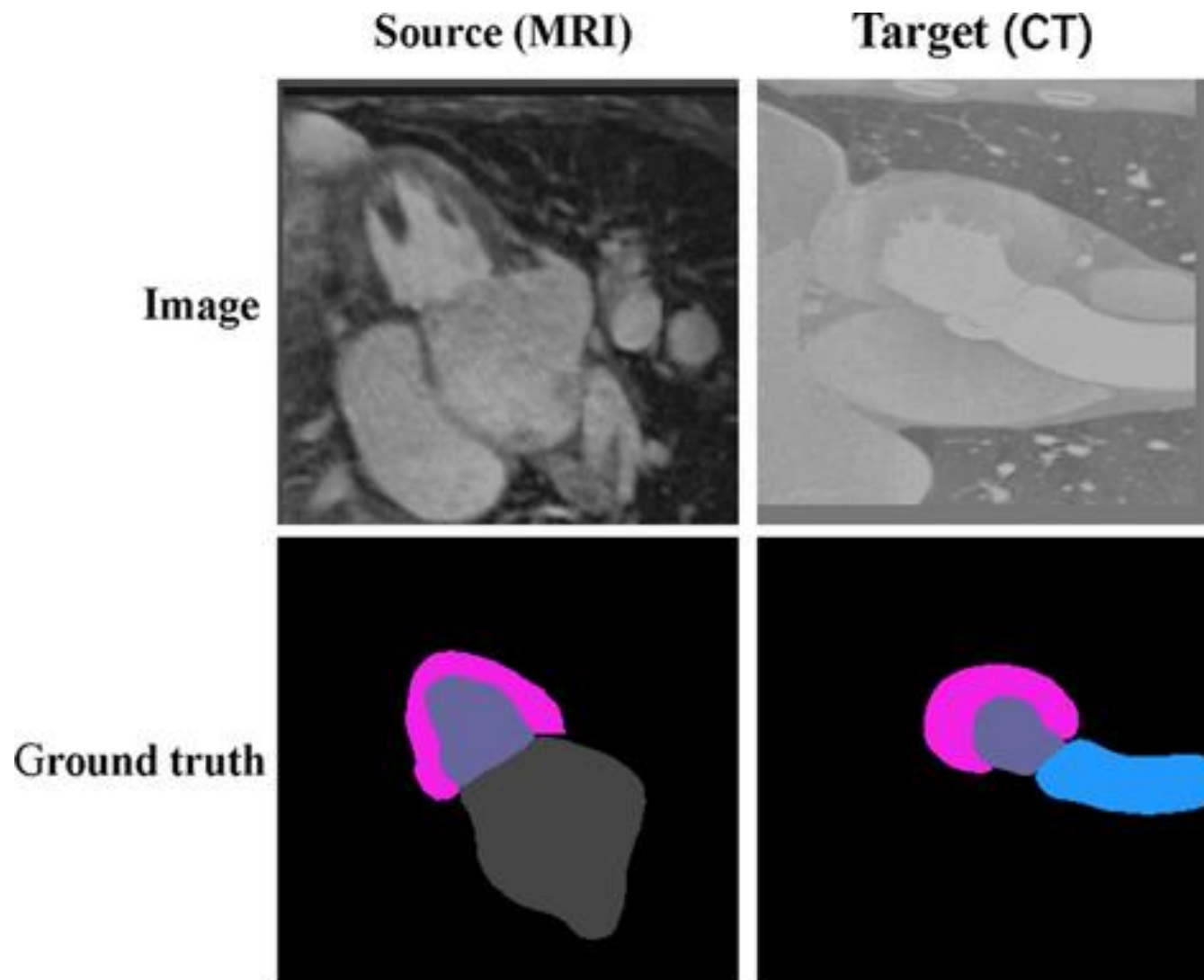
DC-GAD



- Leverage AIM-HIGH annotated data to create a segmentation model for the data being collected at CARDIS study

# Different Technologies

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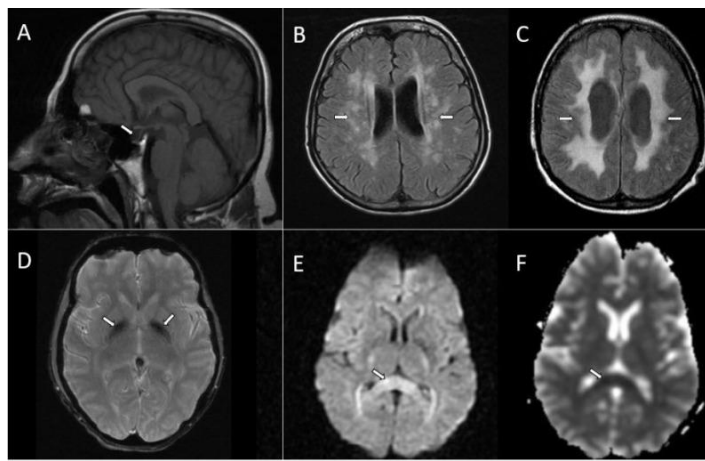
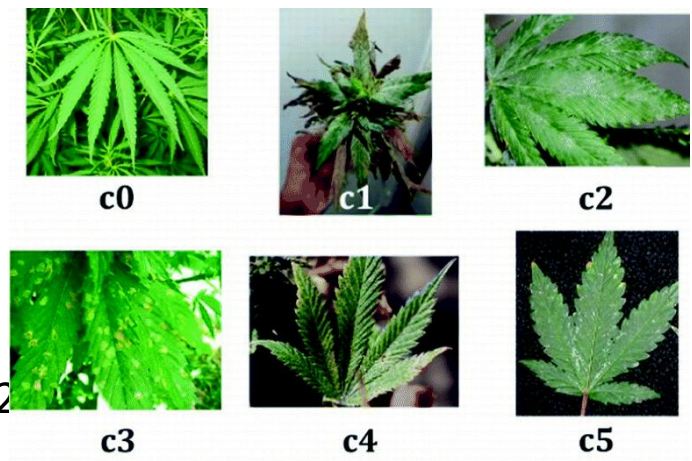
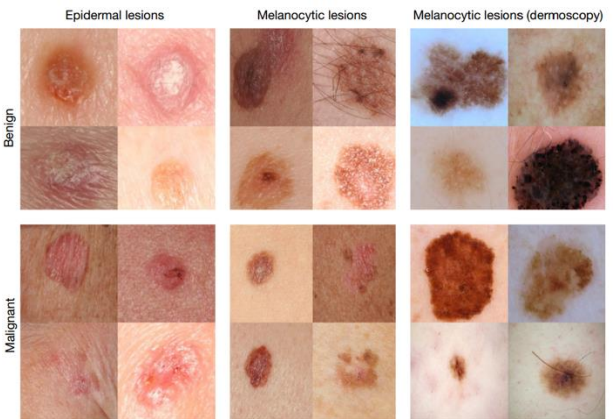
# Degree of Domain Shift

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- **Degree of domain shift** is a measure of how much the distributions of the source and target domains are different
- Previous studies have revealed that the test error generally increases in proportion to the degree of domain shift.

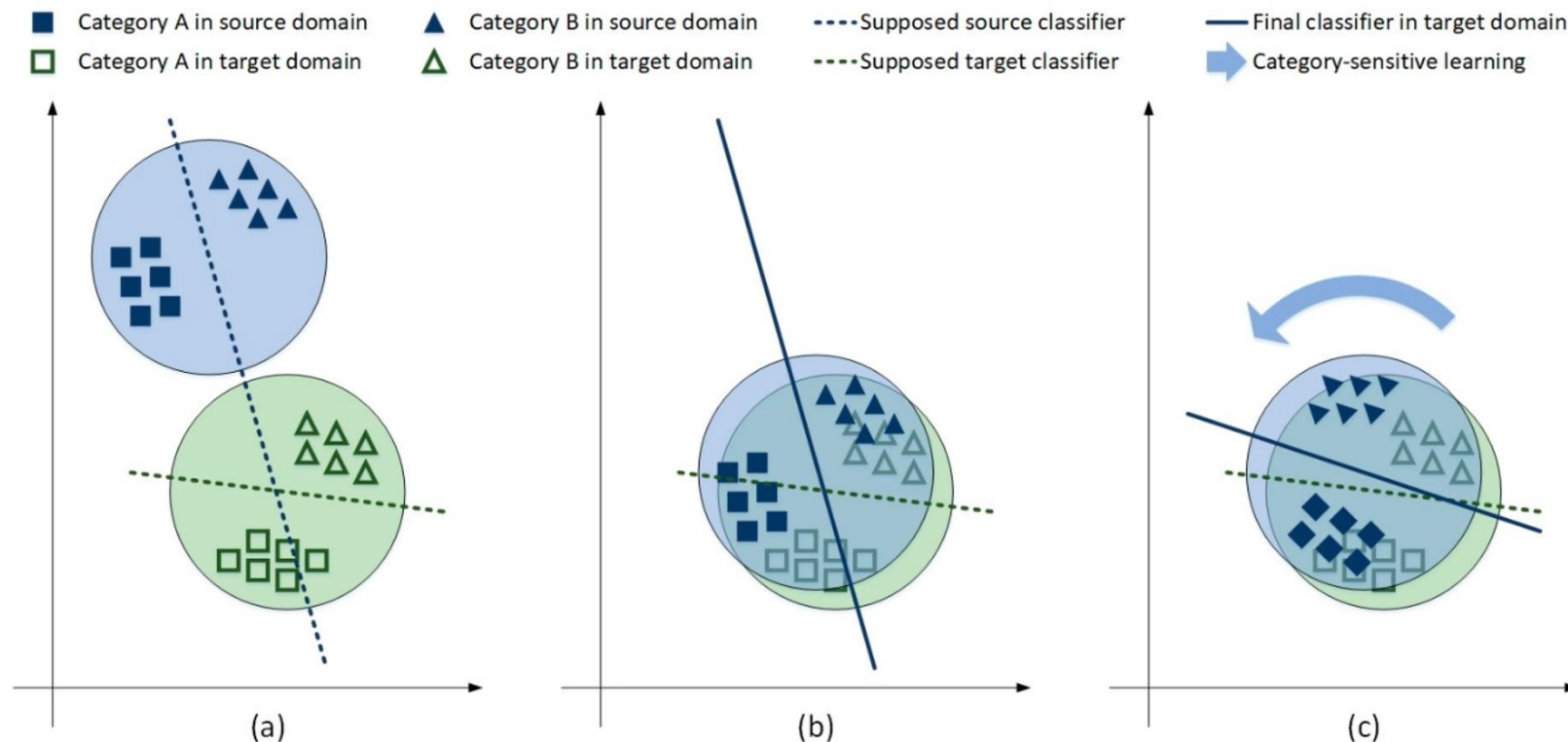
# Degree of Domain Shift

Small or large shift?

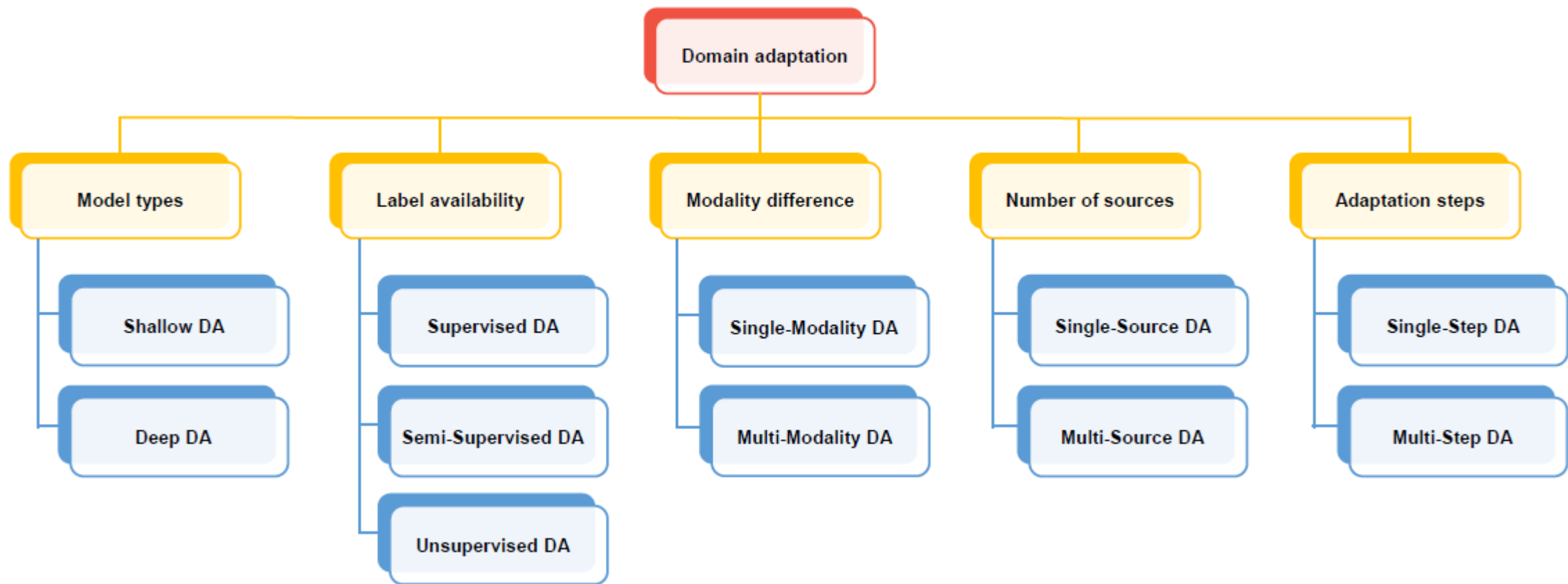


# Domain Adaptation

- **Domain adaptation:** domain adaptation refers to adapting a model trained in one or more source domains to a different one or more target domains.



# Domain Adaptation Categories

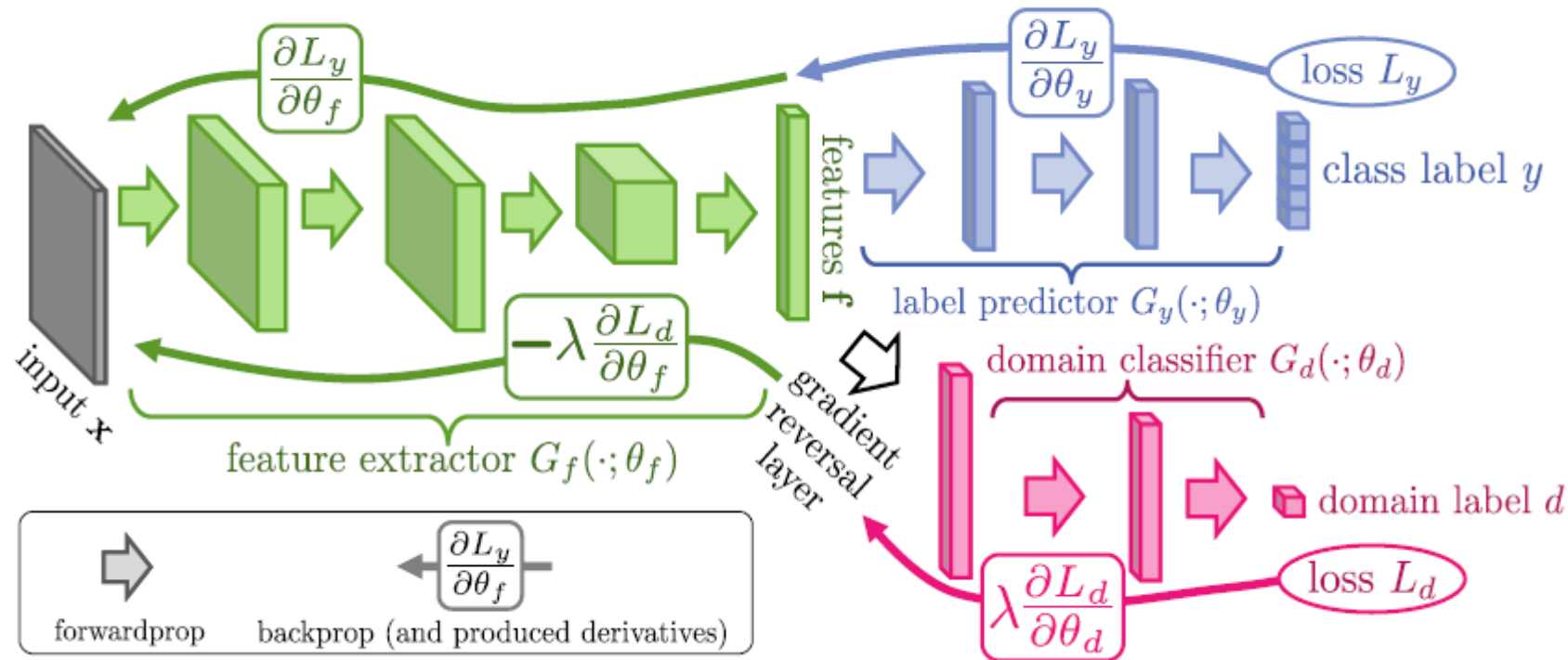


# Supervised Domain Adaptation

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- Essentially transfer learning
  - Fine-tune all layers
  - Fine-tune initial layers
  - Fine-tune final layers

# Domain-Adversarial Training of Neural Networks (Unsupervised)



\*Ganin et al., JMLR, 2016

$$E(\theta_f, \theta_y, \theta_d) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}_y^i(\theta_f, \theta_y) - \lambda \left( \frac{1}{n} \sum_{i=1}^n \mathcal{L}_d^i(\theta_f, \theta_d) + \frac{1}{n'} \sum_{i=n+1}^N \mathcal{L}_d^i(\theta_f, \theta_d) \right)$$

# Summary

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- Domain shift is a problem in machine learning that can make model performance decrease when there is a distribution shift in the test data
- Domain adaptation techniques can be used to make models more robust to data domain shifts

# Thank you!

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